Bureau International des Poids et Mesures

Le système international d'unités
The International System of Units

9 édition 2019

Annexe 2
Annex 2

Recommended Values of Standard Frequencies for Applications Including the Practical Realization of the Metre and Secondary Representations of the Definition of the Second: Rubidium ()

CCL-CCTF Frequency Standards Working Group



Bureau International
des Poids et Mesures

Recommended Values of Standard Frequencies for Applications Including the Practical Realization of the Metre and Secondary Representations of the Definition of the Second: Rubidium ()

CCL-CCTF Frequency Standards Working Group

9th edition 2019

Appendix 2

1. Recommended value [1] of the frequency:

$f\left({}^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}}\right) = 384\ 227\ 981.9\ \mathrm{MHz}$

equivalent to

$\lambda\left({}^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}}\right) = 780\ 246\ 291.6\ \mathrm{fm},$

with an estimated relative standard uncertainty of 5×10^{-10} applies to the radiation of a tunable Extended Cavity Diode Laser, stabilized to the d/f saturated absorption line in a well characterized magnetically shielded rubidium absorption cell by the third-harmonic technique,

- cell temperature (22.5 ± 1) °C
- frequency modulation width, peak-to-peak, (5 ± 1) MHz;
- saturating beam intensity of (120 ± 20) mW cm^{-2}

2. Source data

Adopted value $f = 384\ 227\ 981.9$ MHz
$u_c/y = 5 \times 10^{-10}$

calculated from

$f\left({}^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}}\right)$/ kHz	u_c/y	source data
384 227 981 877.3 (5.5)	1.5×10^{-11}	[2]
384 227 981 867.8 (165)	4.3×10^{-10}	[3]

as a weighted mean. The fractional uncertainty was evaluated to be approximately 4.3×10^{-10} for a 0.01 s averaging time [3]. For the recommendation this uncertainty was rounded to 5×10^{-10}.

^{87}Rb atom, crossover transition between the d and f hyperfine components of the saturated absorption at 780 nm (D2 transition)

References

[1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies http://www.bipm.org/jsp/en/CIPMRecommendations.jsp

[2] J. Ye, S. Swartz, P. Jungner, and J. L. Hall, “Hyperfine structure and absolute frequency of the ^{87}Rb $5P_{3/2}$ state”, *Opt. Lett.* **21**, 1280 (1996).

[3] Y. Bitou, K. Sasaki, H. Inaba, F.-L. Hong, A. Onae, “Rubidium-stabilized diode laser for high-precision interferometer”, *Opt. Eng.* **43**, 900 (2004).